§3.3 Screws: a geometric description of twists

Read introduction to §3.3.

Definition of screw motion

A screw S consists of an axis l, a pitch h, and a magnitude M. A screw motion represents rotation by an amount $\theta = M$ about the axis l followed by translation by an amount $h\theta$ parallel to the axis l. If $h = \infty$ then the corresponding screw motion consists of a pure translation along the axis of the screw by a distance M.

(Be careful: in general ME, talking about screws, the lead of a screw is $2\pi h$, that is the axial direction distance the nut moves with one revolution of the screw relative to the nut. For a single thread screw, though, the lead is also called the pitch, which is not the same as h here; see Shigley p. 422.)

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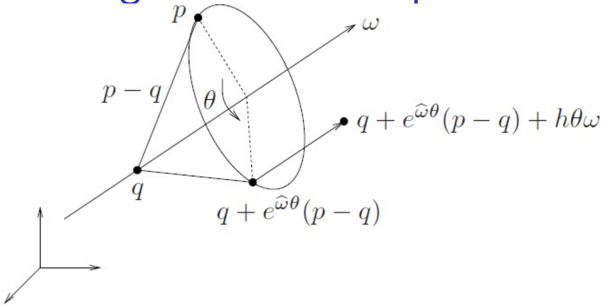

MLS Figure 2.8

$$l = \{q + \lambda\omega : \lambda \in \mathbb{R}\} \qquad (2.39)$$

Rigid body transformation associated with screw S, $p \in \mathbb{R}^3$:

$$gp = q + e^{\hat{\omega}\theta}(p - q) + h\omega\theta$$

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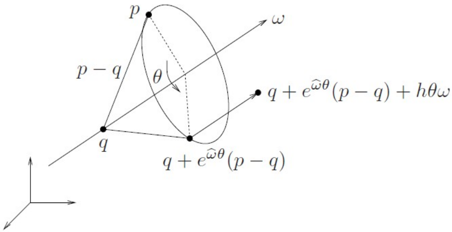

Rigid body transformation associated with screw S, $p \in \mathbb{R}^3$:

$$gp = q + e^{\hat{\omega}\theta}(p - q) + h\omega\theta$$

Homogeneous form:

$$g \begin{bmatrix} p \\ 1 \end{bmatrix} = \begin{bmatrix} e^{\hat{\omega}\theta} & (I - e^{\hat{\omega}\theta})q + h\omega\theta \\ 0 & 1 \end{bmatrix} \begin{bmatrix} p \\ 1 \end{bmatrix}$$

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$$g\begin{bmatrix} p \\ 1 \end{bmatrix} = \begin{bmatrix} e^{\hat{\omega}\theta} & (I - e^{\hat{\omega}\theta})q + h\omega\theta \\ 0 & 1 \end{bmatrix}\begin{bmatrix} p \\ 1 \end{bmatrix}$$

The rigid motion g given by the screw is

$$g = \begin{bmatrix} e^{\hat{\omega}\theta} & (I - e^{\hat{\omega}\theta})q + h\omega\theta \\ 0 & 1 \end{bmatrix} \qquad (2.40)$$

- Maps points attached to the RB from initial ($\theta = 0$) to final coordinates.
- All points specified relative to fixed frame A.

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$$g = \begin{bmatrix} e^{\hat{\omega}\theta} & (I - e^{\hat{\omega}\theta})q + h\omega\theta \\ 0 & 1 \end{bmatrix} \qquad (2.40)$$

$$e^{\hat{\xi}\theta} = \begin{bmatrix} e^{\hat{\omega}\theta} & (I - e^{\hat{\omega}\theta})\hat{\omega}v + \omega\omega^T v\theta \\ 0 & 1 \end{bmatrix} \qquad (2.36)$$

In the line just below the unnumbered equation following eqn. (2.40) (p. 47), MLS write " ... if we choose $v = -\hat{\omega}q + h\omega$, then $\xi = (v, \omega)$ generates the screw motion in equation (2.40) (assuming $||\omega|| = 1, \theta \neq 0$)."

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To show that this is true, we need to show that the 3×1 upper-right sub-matrix in this equation is the same as the 3×1 upper-right sub-matrix in equation (2.40), for $v = -\hat{\omega}q + h\omega$, or

$(I - e^{\hat{\omega}\theta})\hat{\omega}v + \omega\omega^T v\theta = (I - e^{\hat{\omega}\theta})q + h\theta\omega$ for $v = -\hat{\omega}q + h\omega$

Rotating the axis of rotation?

Euler's Theorem states that any orientation $R \in SO(3)$ is equivalent to a rotation about a fixed axis $\omega \in \mathbb{R}^3$ through an angle $\theta \in [0, 2\pi)$ (MLS p. 31). In this statement ω is a unit vector. What happens if this vector ω is rotated through the rotation defined by R?

$$\omega = e^{\hat{\omega}\theta}\omega$$

This is confirmed by the MATLAB script shown in the file "rot_of_axis.pdf".

Proof of statement

$$(I - e^{\hat{\omega}\theta})\hat{\omega}v + \omega\omega^T v\theta = \left[(I - e^{\hat{\omega}\theta})\hat{\omega} + \omega\omega^T\theta\right] v$$

Substituting $v = -\hat{\omega}q + h\omega$ into this, renders

$$\left[(I - e^{\hat{\omega}\theta})\hat{\omega} + \omega\omega^T\theta\right](-\hat{\omega}q + h\omega) = \beta$$

where the dummy variable β can also be written as

$$\beta = -(I - e^{\hat{\omega}\theta})\hat{\omega}^2 q - \omega\omega^T\hat{\omega}q\theta + (I - e^{\hat{\omega}\theta})\hat{\omega}\omega h + \omega\omega^T\omega h\theta$$

The 3rd term in the above expression for β is zero, as $\hat{\omega}\omega \equiv 0$. Therefore

$$\beta = -(I - e^{\hat{\omega}\theta})\hat{\omega}^2 q - \omega\omega^T\hat{\omega}q\theta + \omega\omega^T\omega h\theta$$

Proof of statement

$$\beta = -(I - e^{\hat{\omega}\theta})\hat{\omega}^2 q - \omega\omega^T \hat{\omega} q\theta + \omega\omega^T \omega h\theta$$

Consider now only the 1st term in β above. From Lemma 2.3 p. 28, for $||\omega|| = 1 : \hat{\omega}^2 = \omega\omega^T - I$. Therefore

$$\begin{aligned}
-(I - e^{\hat{\omega}\theta})\hat{\omega}^2 q &= -(I - e^{\hat{\omega}\theta})(\omega\omega^T - I)q \\
&= (I - e^{\hat{\omega}\theta})q - (I - e^{\hat{\omega}\theta})\omega\omega^T q \\
&= (I - e^{\hat{\omega}\theta})q - (\omega\omega^T - e^{\hat{\omega}\theta}\omega\omega^T)q \\
&= (I - e^{\hat{\omega}\theta})q - (\omega\omega^T - \omega\omega^T)q \\
&= (I - e^{\hat{\omega}\theta})q
\end{aligned}$$

$$\beta = (I - e^{\hat{\omega}\theta})q - \omega\omega^T \hat{\omega} q\theta + \omega\omega^T \omega h\theta$$

Proof of statement

$$\beta = (I - e^{\hat{\omega}\theta})q - \omega\omega^T\hat{\omega}q\theta + \omega\omega^T\omega h\theta$$

Consider now the last two terms in β above. Remembering that $\hat{\omega}^2\omega = \hat{\omega}\hat{\omega}\omega = 0$, and that from Lemma 2.3 p. 28, for $||\omega|| = 1 : \hat{\omega}^2 = \omega\omega^T - I$ and $\hat{\omega}^3 = -\hat{\omega}$, it is easy to show that

$$\begin{aligned}
\omega\omega^T\omega h\theta - \omega\omega^T\hat{\omega}q\theta &= (\hat{\omega}^2 + I)\omega h\theta - (\hat{\omega}^2 + I)\hat{\omega}q\theta \\
&= \omega h\theta - (\hat{\omega}^3 + \hat{\omega})q\theta \\
&= \omega h\theta
\end{aligned}$$

$$\beta = (I - e^{\hat{\omega}\theta})q + \omega h\theta$$

which is recognized as the 3×1 upper-right sub-matrix in equation (2.40)

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Conclusion: the screw S with axis l, pitch h and magnitude M:

$$g = \begin{bmatrix} e^{\hat{\omega}\theta} & (I - e^{\hat{\omega}\theta})q + h\omega\theta \\ 0 & 1 \end{bmatrix}$$

is described by the twist $\xi = (v, \omega) = [\ v^T \quad \omega^T\ ]^T$

where, with the displacement vector $\vec{d}$,

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Twist $\xi = (v, \omega)$, where, with the displacement vector $\vec{d}$,

Motion includes rotation	Translation only
$\|\|\omega\|\| = 1$	$\|\|\omega\|\| = 0$
$v = -\hat{\omega} q + h\omega$	$\|\|v\|\| = 1$
l aligned with $\vec{\omega}$	l aligned with $\vec{d}$ or $\vec{v}$
$M = \theta$, angle of rotation (rad)	$M = \theta = \|\vec{d}\|$
$\vec{d} = h\theta\vec{\omega}$	$\vec{d} = M\vec{v}$; $h = \infty$
g as in eqn. (2.40)	g as in eqn. (2.41)

$$g = \begin{bmatrix} I & \theta v \\ 0 & 1 \end{bmatrix} = e^{\hat{\xi}\theta} \text{ for } \xi = (v, 0) \quad (2.41)\&(2.32)$$

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We have shown that any screw can be expressed as a twist.

We will now also show that all twists can be expressed as screws.

Let $\hat{\xi} \in se(3)$ be a twist with coordinates $\xi = (v, \omega) \in \mathbb{R}^6$. No assumption $||\omega|| = 1$ and allowing both pure rotation and pure translation.

1. Pitch: $h = \dfrac{\omega^T v}{||\omega||^2}$ (2.42)
2. Axis:

$$l = \begin{cases} \left\{ \dfrac{\hat{\omega} v}{||\omega||^2} + \lambda\omega : \lambda \in \mathbb{R} \right\}, & \text{if } \omega \neq 0 \\ \{0 + \lambda v : \lambda \in \mathbb{R}\}, & \text{if } \omega = 0 \end{cases} \qquad (2.43)$$

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Let $\hat{\xi} \in se(3)$ be a twist with coordinates $\xi = (v, \omega) \in \mathbb{R}^6$. No assumption $||\omega|| = 1$ and allowing both pure rotation and pure translation.

1. Pitch: $h = \dfrac{\omega^T v}{||\omega||^2}$ $\quad (2.42)$

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3. Magnitude (note, different from MLS eq. (2.44)):

$$M = \begin{cases} \theta, & \text{if } \omega \neq 0 \\ ||d||, & \text{if } \omega = 0 \end{cases}$$

indicating net rotation if $\omega \neq 0$, net translation otherwise.

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Exploring $h = \dfrac{\omega^T v}{||\omega||^2}$ and $l = \dfrac{\hat{\omega} v}{||\omega||^2} + \lambda\omega$ (implied $||\omega|| \neq 1$):

$$v = -\hat{\omega} q + h\omega$$

$$\begin{aligned}
\omega^T v &= -\omega^T \hat{\omega} q + h\omega^T\omega \\
&= \omega^T \hat{\omega}^T q + h||\omega||^2 \\
&= (\hat{\omega}\omega)^T q + h||\omega||^2 \\
&= h||\omega||^2
\end{aligned}$$

Thus, if $v = -\hat{\omega} q + h\omega$, $\qquad h = \dfrac{\omega^T v}{||\omega||^2}$.

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Exploring $h = \dfrac{\omega^T v}{||\omega||^2}$ and $l = \dfrac{\hat{\omega} v}{||\omega||^2} + \lambda\omega$:

Note that, for a unit vector $\vec{\omega}$ and another vector $\vec{\tau}$ in 3D space, one can show that

$$(\vec{\omega} \times \vec{\tau}) \times \vec{\omega} + (\vec{\omega} \cdot \vec{\tau})\vec{\omega} = \vec{\tau}$$

or, if $\vec{\omega}$ and $\vec{\tau}$ are written in component form as the two 3×1 column vectors ω and τ, respectively,

$$\widehat{(\hat{\omega}\tau)}\,\omega + (\omega^T\tau)\omega = \tau\ .$$

(See MATLAB code file “ident4MLS_Theor2_17.pdf”)

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Exploring $h = \dfrac{\omega^T v}{||\omega||^2}$ and $l = \dfrac{\hat{\omega} v}{||\omega||^2} + \lambda\omega$:

Remember earlier definition of l?

Consider a twist $\xi = (v, \omega)$ where there is no requirement that $||\omega|| = 0$. Let $q = \dfrac{\hat{\omega} v}{||\omega||^2}$ and $h = \dfrac{\omega^T v}{||\omega||^2}$.Then, let

$$
\begin{aligned}
a &= -\hat{\omega} q + h\omega = \hat{q}\omega + h\omega \\
&= \widehat{\left(\frac{\hat{\omega} v}{||\omega||^2}\right)}\omega + \frac{\omega^T v}{||\omega||^2}\,\omega
\end{aligned}
$$

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Exploring $h = \dfrac{\omega^T v}{||\omega||^2}$ and $l = \dfrac{\hat{\omega} v}{||\omega||^2} + \lambda\omega$:

$$a = -\hat{\omega} q + h\omega = \widehat{\left(\frac{\hat{\omega} v}{||\omega||^2}\right)}\omega + \frac{\omega^T v}{||\omega||^2}\,\omega = -\hat{\omega}\frac{\hat{\omega} v}{||\omega||^2} + \frac{\omega^T v}{||\omega||^2}\,\omega$$

Let $\underline{\omega} = \omega/||\omega||$, that is, a unit vector in the direction of ω.

$$a = \widehat{(\underline{\hat{\omega}} v)}\underline{\omega} + (\underline{\omega}^T v)\ \underline{\omega} = v$$

Conclusion: the twist $\xi = (v, \omega)$, $||\omega|| \neq 1$, does in fact correspond to a screw, as $v = -\hat{\omega} q + h\omega$, with

$$q = \frac{\hat{\omega} v}{||\omega||^2} \ \ , \ \ h = \frac{\omega^T v}{||\omega||^2} \ \ , \ \ l = \frac{\hat{\omega} v}{||\omega||^2} + \lambda\omega$$

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Study Proposition 2.10 p. 48 and take note of the definition of the concept *unit twist* 1st paragraph on p. 49.

Theorem 2.11 (Chasles). Every rigid body motion can be realized by a rotation about an axis combined with a translation parallel to that axis.

Study the rest of §3.3 including example 2.3.

Summary following from Chasles

Every rigid body motion can be expressed as a screw, that is, transformation $g_{ab} \in SE(3)$ that generates a body fixed axis system frame B, rotated relative to the inertial frame A, by rotation matrix R_{ab}, and with origin at coordinates p_{ab} in frame A,

$$g_{ab} = \begin{bmatrix} R_{ab} & p_{ab} \\ 0 & 1 \end{bmatrix} .$$

By proposition 2.9 p. 42 there exists a twist $\xi = [\ v \quad \omega\ ]^T$ such that $g_{ab} = e^{\hat{\xi}\theta}$, and v, ω and θ can be determined if R_{ab} and p_{ab} are known.

In this it is possible to have $||\omega|| \neq 1$ in the case where $||\omega|| \neq 0$. With ξ known, the screw associated with this twist can be calculated using eqn.s (2.42) & (2.43).